An Introduction to Counting and Probability

Ann Zhou

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1 Introduction

pov me when I have the mental capacity of a toddler trying to put the square block into the triangle hole... If there are any incorrect answers or concepts, I will probably catch it in the near future. I wrote this at midnight, and lord knows I shouldn't be deemed trustworthy past my bedtime.

2 Principles of Counting

2.1 Fundamental Counting Principle

- **What it is:** A rule used to find the total number of outcomes when making multiple independent choices in sequence.
- **Equations/Definitions:** If there are a ways to make a first choice, b ways to make a second choice, and c ways to make a third choice, the total number of unique sequences of choices is:

$$\text{Total Outcomes} = a \times b \times c \times \dots$$

- **How/What it is used for:** Breaking down a complex tracking or arrangement problem into distinct slots or stages.
- **Tricks & Common Mistakes:**
  - *The "And" vs. "Or" Trap:* Multiply choices when performing a sequence of actions (**and**). Add choices when selecting between mutually exclusive alternatives (**or**).
- **Example Question:** A menu offers 3 appetizers, 5 main courses, and 4 desserts. How many different 3-course meals can be created?
  Solution: Using the principle, multiply the independent choices together: $3 \times 5 \times 4 = 60$.

2.2 Permutations

- **What it is:** The number of unique ways to arrange a specific subset of items where **order matters**.
- **Equations/Definitions:** The number of ways to arrange r items selected from a total pool of n distinct items:

$$P(n, r) = \frac{n!}{(n-r)!}$$

- **How/What it is used for:** Standard linear seatings, awarding gold/silver/bronze medals, or creating specific alphanumeric passwords.
- **Tricks & Common Mistakes:**
  - *Overcounting Selections:* Make sure order actually matters (e.g., choosing a President and Vice President vs. just choosing a 2-person committee).
- **Example Question:** How many unique 3-letter codes can be formed from the letters in the word `NUMBER` if no letter can be repeated?
  Solution: There are 6 unique letters, and we want to arrange 3 of them. $P(6,3) = \frac{6!}{(6-3)!} = 6 \times 5 \times 4 = 120$.

2.3 Combinations

- **What it is:** The number of unique ways to select a subset of items where **order does not matter**.
- **Equations/Definitions:** The number of ways to choose r items from a total pool of n distinct items:
$$C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$
- **How/What it is used for:** Selecting committees, forming hands of cards, or identifying matching paths.
- **Tricks & Common Mistakes:**
  - *Symmetry Identity:* Remember that $\binom{n}{r} = \binom{n}{n-r}$. Choosing who to leave out is structurally the exact same as choosing who to include.
- **Example Question:** A math team consists of 8 students. How many ways can a 3-person delegation be chosen to attend a meet?
  Solution: Since the roles within the delegation are identical, order does not matter. Use combinations: $\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$.

2.4 Multi-sets (Permutations with Repetition)

- **What it is:** Finding the number of distinct linear arrangements when some objects in the pool are completely identical to one another.
- **Equations/Definitions:** If there are N total objects where object 1 repeats n_1 times, object 2 repeats n_2 times, etc., the number of unique arrangements is:
$$\text{Arrangements} = \frac{N!}{n_1! \times n_2! \times \cdots \times n_k!}$$
- **How/What it is used for:** Rearranging the letters of words containing duplicate letters or grid path walks.
- **Tricks & Common Mistakes:**

- *Forgetting the Denominator:* If you treat identical objects as distinct, you drastically overcount the configurations. Always divide out by the factorials of the repeated groups.

- **Example Question:** How many distinct ways can the letters of the word `PROBABILITY` be rearranged?
Solution: There are 11 total letters. The letter `B` appears 2 times, and the letter `I` appears 2 times. The total number of arrangements is $\frac{11!}{2! \times 2!} = \frac{39{,}916{,}800}{4} = 9,979,200$.

2.5 Stars & Stripes (a.k.a. Sticks & Stones / Stars & Bars)

- **What it is:** A visual counting technique used to distribute identical items into distinct groups.

- **Equations/Definitions:**

  - To distribute n identical items into k distinct groups where groups **can be empty** (non-negative integers):
$$\text{Ways} = \binom{n+k-1}{k-1}$$
  - To distribute n identical items into k distinct groups where each group **must have at least one item** (positive integers):
$$\text{Ways} = \binom{n-1}{k-1}$$

- **How/What it is used for:** Finding the number of integer solutions to equations like $x_1 + x_2 + x_3 = 10$, or distributing identical pieces of candy to children.

- **Tricks & Common Mistakes:**

  - *Misidentifying Items vs. Bins:* Make sure the items being distributed are completely *identical* (stars) and the receivers are *distinct* (separating lines).

- **Example Question:** How many non-negative integer solutions exist for the equation $x + y + z = 7$?
Solution: Here, we have $n = 7$ identical units (stars) and $k = 3$ variables (bins). The number of separating lines needed is $k - 1 = 2$. Applying the formula: $\binom{7+3-1}{3-1} = \binom{9}{2} = \frac{9 \times 8}{2 \times 1} = 36$.

3 Probability Theory

3.1 Unions and Mutually Exclusive Events

- **What it is:** Calculating the probability that at least one of multiple events occurs.

- **Equations/Definitions:**

  - *General Principle (Inclusion-Exclusion):* $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
  - *Mutually Exclusive Events:* If A and B cannot happen at the same time, then $P(A \cap B) = 0$, simplifying to:
$$P(A \cup B) = P(A) + P(B)$$

- **How/What it is used for:** Calculating probability profiles when conditions overlap or are explicitly separated.
- **Tricks & Common Mistakes:**
  - *Double Counting the Intersection:* If events can happen simultaneously, you must subtract their overlap. Forgetting to subtract $P(A \cap B)$ is the most common error.
- **Example Question:** If you roll a single fair 6-sided die, what is the probability that the result is an even number or a multiple of 3?
  Solution: Let A be rolling an even number $\{2, 4, 6\}$, so $P(A) = \frac{3}{6}$. Let B be rolling a multiple of 3 $\{3, 6\}$, so $P(B) = \frac{2}{6}$. The intersection $A \cap B$ is $\{6\}$, so $P(A \cap B) = \frac{1}{6}$. Thus, $P(A \cup B) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3}$.

3.2 Intersections and Independence

- **What it is:** Finding the probability that multiple events all occur simultaneously or sequentially.
- **Equations/Definitions:**
  - *General Multiplication Rule:* $P(A \cap B) = P(A) \times P(B \mid A)$
  - *Independent Events:* If the occurrence of one event does not change the likelihood of the other, then $P(B \mid A) = P(B)$, which yields:

$$P(A \cap B) = P(A) \times P(B)$$

- **How/What it is used for:** Successive coin flips, rolling multiple dice, or tracking continuous multi-stage systems.
- **Tricks & Common Mistakes:**
  - *Blindly Multiplying:* Never assume independence without verifying. If you pull cards from a deck *without replacement*, the probabilities change at every single stage.
- **Example Question:** A coin is flipped and a fair 6-sided die is rolled. What is the probability of getting heads and rolling a 5?
  Solution: The coin flip and die roll are completely independent events. $P(\text{Heads}) = \frac{1}{2}$ and $P(\text{Rolling a 5}) = \frac{1}{6}$. Therefore, $P(\text{Heads} \cap 5) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.

3.3 Complements

- **What it is:** The probability of an event *not* occurring.
- **Equations/Definitions:** Let A^c or A' denote the complement of event A:

$$P(A^c) = 1 - P(A)$$

- **How/What it is used for:** Drastically simplifying direct counting problems, particularly when asked to find the probability of "at least one" outcome occurring.
- **Tricks & Common Mistakes:**

- *Misinterpreting "At Least One":* The mathematical complement of "at least one success" is "exactly zero successes".

- **Example Question:** If you flip a fair coin 4 times, what is the probability of getting at least one head?
  Solution: Calculating 1, 2, 3, or 4 heads directly requires adding multiple cases. Instead, calculate the complement (getting zero heads, meaning all tails): $P(\text{All Tails}) = \left(\frac{1}{2}\right)^4 = \frac{1}{16}$. Subtract this value from 1: $P(\text{At Least One Head}) = 1 - \frac{1}{16} = \frac{15}{16}$.

3.4 Conditional Probability

- **What it is:** The probability of an event occurring given that another event has already occurred.

- **Equations/Definitions:** The probability of event A occurring given that event B has occurred is defined as:
$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) > 0$$

- **How/What it is used for:** Restricting the available sample space based on known contextual constraints.

- **Tricks & Common Mistakes:**

  - *Confusing $P(A \mid B)$ with $P(B \mid A)$:* Order matters immensely. The probability that it is cloudy given that it is raining is 100%, but the probability that it is raining given that it is cloudy is much lower.

- **Example Question:** A family has two children. Given that at least one of the children is a boy, what is the probability that both children are boys? (Assume boys and girls are equally likely).
  Solution: The full sample space for two children is $\{BB, BG, GB, GG\}$. The condition "at least one is a boy" completely eliminates GG, narrowing our working sample space down to 3 outcomes: $\{BB, BG, GB\}$. Within this new restricted space, only 1 outcome (BB) features two boys. Thus, the probability is $\frac{1}{3}$.

3.5 Bayes's Theorem

- **What it is:** An algebraic formula used to update existing probability assessments when new evidence emerges. It effectively reverses conditional dependencies.

- **Equations/Definitions:**
$$P(A \mid B) = \frac{P(B \mid A) \times P(A)}{P(B)}$$
  Using the Law of Total Probability for the denominator:
$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c)$$

- **How/What it is used for:** Interpreting medical testing accuracy, quality control diagnostics, and updating beliefs based on a test result.

- **Tricks and Common Mistakes:**
  - *Ignoring the Base Rate:* People often look only at a test's individual accuracy while ignoring how rare the condition actually is in the baseline population.
- **Example Question:** A disease affects 1% of a population. A diagnostic test is 90% accurate (meaning it yields a positive result 90% of the time if someone has it, and a negative result 90% of the time if they don't). If a person tests positive, what is the probability they actually have the disease?
  Solution: Let D be having the disease, and $+$ be testing positive. We want to find $P(D \mid +)$.
  - $P(D) = 0.01$, so $P(D^c) = 0.99$.
  - $P(+ \mid D) = 0.90$ (True Positive rate).
  - $P(+ \mid D^c) = 0.10$ (False Positive rate).
  - Calculate total probability of testing positive: $P(+) = (0.90 \times 0.01) + (0.10 \times 0.99) = 0.009 + 0.099 = 0.108$.
  - Apply Bayes's Theorem: $P(D \mid +) = \frac{0.009}{0.108} = \frac{9}{108} = \frac{1}{12} \approx 8.33\%$.

3.6 Expected Value

- **What it is:** The long-run theoretical average outcome of a random variable across many repeated trials.
- **Equations/Definitions:** For a discrete random variable X with outcomes x_i and corresponding probabilities p_i:

$$E(X) = \sum x_i p_i = x_1 p_1 + x_2 p_2 + \cdots + x_n p_n$$

- **How/What it is used for:** Evaluating game fairness, managing insurance premiums, and determining optimal mathematical strategy.
- **Tricks and Common Mistakes:**
  - *Linearity of Expectation:* $E(X + Y) = E(X) + E(Y)$ is always true, even if variables X and Y are totally dependent on one another. This fact is a major shortcut in competition math.
- **Example Question:** A game costs \$2 to play. A player rolls a fair 6-sided die. If they roll a 6, they win \$6; if they roll any other number, they win nothing. What is the expected net profit/loss per game?
  Solution: The net outcome values are: rolling a 6 yields $+\$4$ (profit of $\$6 - \$2$), and rolling a 1–5 yields $-\$2$. The expected value is: $E(X) = \left(4 \times \frac{1}{6}\right) + \left(-2 \times \frac{5}{6}\right) = \frac{4}{6} - \frac{10}{6} = -\frac{6}{6} = -\$1$. You expect to lose \$1 on average per game.

4 Reference Material & Toolkits

4.1 Features of a Standard 52-Card Deck

When dealing with contest problems, missing structural card assumptions will break your calculations instantly. Double-check your setup with this baseline:

Table 1: Composition of a Standard Deck

Category	Size / Composition	Details
Total Cards	**52**	Divided evenly into 4 suits of 13 cards each.
Colors	**26 Black, 26 Red**	Clubs/Spades are Black; Diamonds/Hearts are Red.
Suits	**4 Suits**	Clubs (♣), Diamonds (♢), Hearts (♡), Spades (♠).
Ranks	**13 Ranks per Suit**	Ace (A), 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack (J), Queen (Q), King (K).
Face Cards	**12 Total**	Jacks, Queens, and Kings only. (**Aces are not face cards**).

4.2 Prime Factorizations and Fraction Reduction

- **What it is:** Expressing numbers as prime factor bundles to cleanly cancel out terms without doing tedious long division.
- **Equations/Definitions:** Every integer $n > 1$ can be uniquely written as:

$$n = p_1^{e_1} \cdot p_2^{e_2} \dots p_k^{e_k}$$

- **How/What it is used for:** Rapidly simplifying probability fractions that feature massive combinations or factorials.
- **Tricks and Common Mistakes:**
  - *Expanding Before Canceling:* Never multiply out big factorials or combinations in a probability fraction. Factor them into prime powers first, then subtract exponents to reduce them.
- **Example Question:** Reduce the fraction $\frac{\binom{10}{4}}{105}$ to its simplest form.
  Solution: First, break down the numerator: $\binom{10}{4} = \frac{10\times9\times8\times7}{4\times3\times2\times1} = 10 \times 3 \times 7 = 210$.
  Write both numerator and denominator in their prime factor forms:
  - $210 = 2^1 \times 3^1 \times 5^1 \times 7^1$
  - $105 = 3^1 \times 5^1 \times 7^1$

  Cancel the shared prime blocks ($3 \times 5 \times 7$), leaving exactly $2^1 = 2$.

4.3 Geometric Probability

- **What it is:** A type of probability where likelihoods are calculated based on ratios of continuous geometric measures—such as lengths, areas, or volumes—rather than discrete item counts.
- **Equations/Definitions:** For a continuous sample space, the probability of a successful event is defined as the ratio of the favorable measure to the total measure:

$$P(\text{Event}) = \frac{\text{Favorable Measure}}{\text{Total Measure}}$$

  (where measure refers to length, area, or volume depending on the dimension of the problem).
- **How/What it is used for:** Solving continuous random variable problems, meeting-time window puzzles, and target/dartboard hit simulations.

- **Tricks and Common Mistakes:**
  - *Boundary Neglect:* In continuous spaces, strict inequalities ($<$ or $>$) and non-strict inequalities ($\leq$ or $\geq$) yield the exact same probability because single points and lines have a measure (area/length) of zero.
- **Example Question:** Two friends agree to meet at a coffee shop between 2:00 PM and 3:00 PM. Each person arrives independently at a random time and waits for 15 minutes before leaving. What is the probability that they meet?
  Solution: Let x and y be their arrival times measured in minutes from 2:00 PM ($0 \leq x \leq 60$, $0 \leq y \leq 60$). The total sample space area is $60 \times 60 = 3600$. They meet if $|x - y| \leq 15$, which means $-15 \leq x - y \leq 15$. The region where they *do not* meet consists of two right triangles in the corners of the square, each with legs of length $60 - 15 = 45$. The failure area is $\frac{1}{2}(45)(45) + \frac{1}{2}(45)(45) = 2025$. Thus, the successful meeting area is $3600 - 2025 = 1575$. The probability is: $P(\text{Meet}) = \frac{1575}{3600} = \frac{7}{16}$.

4.4 Binomial Theorem and Binomial Probability

- **What it is:** A formula used to calculate the probability of achieving exactly k successes across n independent yes/no (Bernoulli) trials, where each trial has a constant success probability p.
- **Equations/Definitions:** The binomial probability formula for k successes in n trials is:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

  where $q = 1 - p$ is the probability of failure on any given trial.
- **How/What it is used for:** Finding the likelihood of specific success counts in repeated independent events, such as coin flip sequences, basketball free-throw streaks, or quality control testing.
- **Tricks and Common Mistakes:**
  - *Forgetting the Combination Coefficient:* Remember to include $\binom{n}{k}$ to account for all the different orders/arrangements in which the k successes can occur, not just one specific sequence.
- **Example Question:** A basketball player makes 70% of her free throws. If she shoots 5 free throws, what is the probability that she makes exactly 4 of them?
  Solution: Here, $n = 5$ trials, $k = 4$ successes, success probability $p = 0.7$, and failure probability $q = 0.3$. Applying the binomial formula:
  $P(X = 4) = \binom{5}{4}(0.7)^4(0.3)^1 = 5 \times (0.2401) \times 0.3 = 5 \times 0.07203 = 0.36015$ (or 36.015%).

5 Practice Problems

5.1 Foundational Problems

1. **(HMMT November 2023, General Round)** Four people are playing rock-paper-scissors. They each play one of the three options (rock, paper, or scissors) independently at random,

with equal probability of each choice. Compute the probability that someone beats everyone else.
Answer: $\frac{4}{27}$
Not-So-Brief Hint: Total outcomes are $3^4 = 81$. A single player beats everyone else if that player chooses the winning weapon and the other 3 players all choose the losing weapon. There are 3 choices for the winning weapon and 4 choices for which player wins, giving $3 \times 4 = 12$ successful outcomes. Thus, $\frac{12}{81} = \frac{4}{27}$.

2. **(AMC 12A 2021, Problem 13)** A regional math contest has 10 problems. The score for each problem is 1 or 0. A student's total score is the sum of their scores on the 10 problems. How many possible test papers can be generated such that the student scores exactly 5 points, but no two correct answers are consecutive?
Answer: 6
Not-So-Brief Hint: Choosing 5 non-consecutive correct answers out of 10 is equivalent to placing 5 correct items into the gaps between 5 incorrect items. Use the gap formula $\binom{(10-5)+1}{5} = \binom{6}{5} = 6$.

3. **(HMMT 2013, Combinatorics Test)** A standard 52-card deck contains cards of 4 suits and 13 numbers, with exactly one card for each pairing of suit and number. If Maya draws two cards with replacement from this deck, what is the probability that the two cards have the same suit or have the same number, but not both?
Answer: $\frac{15}{52}$
Not-So-Brief Hint: Let S be the event that both cards share the same suit, and N be the event that they share the same number. Use Principle of Inclusion-Exclusion: $P(S \text{ or } N \text{ but not both}) = P(S)+P(N)-2P(S \text{ and } N)$ where both cards being identical counts towards the intersection.

4. **(AMC 10B 2022, Problem 16)** An ant is at the vertex of a cube. Every second, it randomly moves along one of the 3 edges connected to its current vertex to an adjacent vertex. What is the probability that after 4 seconds, the ant is back at its starting vertex?
Answer: $\frac{7}{27}$
Not-So-Brief Hint: Set up state transitions based on the graph distance from the starting vertex (0 steps away, 1 step away, or 3 steps away diagonally) and compute the transition probabilities step-by-step up to step 4.

5. **(AMC 10A 2020 / AMC 12A 2020, Problem 19)** A coin that lands on heads with probability p is flipped 10 times. The probability of obtaining an even number of heads is exactly $\frac{1}{2}$. What are all possible values of p?
Answer: $\frac{1}{2}$
Not-So-Brief Hint: Use binomial expansions. The probability of an even number of heads can be expressed using the binomial theorem expansion of $\frac{(p+q)^{10}+(q-p)^{10}}{2}$ where $q = 1 - p$, simplifying to an equation solved uniquely at $p = 1/2$.

5.2 Some Harder Problems

1. **(2001 AIME I, Problem 6)** A fair six-sided die is rolled four times. The probability that each of the final three rolls is at least as large as the roll preceding it may be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.
Answer: 79
Not-So-Brief Hint: This is equivalent to counting non-decreasing sequences of length 4 from

numbers 1 through 6. Using Stars and Bars, the number of valid sequences is $\binom{4+6-1}{4} = 126$. Total outcomes are $6^4 = 1296$, yielding $\frac{126}{1296} = \frac{7}{72}$, so $m + n = 79$.

2. **(2001 AIME II, Problem 9)** Each unit square of a 3×3 unit-square grid is colored blue or red with equal probability. Find $m + n$ for probability $\frac{m}{n}$ of obtaining a grid without a 2×2 red square.
Answer: 929
Not-So-Brief Hint: Total possible grids are $2^9 = 512$. Note that there are four possible positions for a 2×2 subgrid. Use the Principle of Inclusion-Exclusion to count the number of grids containing at least one 2×2 red block, then subtract from the total.

3. **(1983 AIME, Problem 7)** Twenty-five knights are seated at a round table, and 3 are chosen at random. Find the sum of the numerator and denominator for the probability that at least two are adjacent.
Answer: 57
Not-So-Brief Hint: Use complementary counting: find the probability that *none* of the 3 chosen knights are adjacent. This is equivalent to choosing 3 non-consecutive items from 25 arranged in a circle, solved by breaking the circle at a fixed choice.

4. **(HMMT November 2017, Guts Round)** A box contains three balls, each of a different color. Every minute, Randall randomly draws a ball from the box, notes its color, and then returns it to the box. Consider the following conditions: (1) Some ball has been drawn at least three times; (2) Every ball has been drawn at least once. What is the probability that condition (1) is met before condition (2)?
Answer: $\frac{4}{9}$
Not-So-Brief Hint: Track the state transitions of distinct colors drawn. To meet condition (1) first, you must draw the same initial ball 3 times in a row before ever drawing either of the other two colors, which happens with probability $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$, plus paths where a second color is drawn once before the first color hits 3, leading to the total sum of $\frac{4}{9}$.

5. **(AMC 12B 2023, Problem 15)** Three real numbers x, y, z are chosen independently and uniformly at random from the interval $[0, 1]$. What is the probability that no two of the numbers differ by more than $\frac{1}{2}$?
Answer: $\frac{1}{4}$
Not-So-Brief Hint: The condition means $\max(x, y, z) - \min(x, y, z) \leq \frac{1}{2}$. Geometrically, this cuts out corner tetrahedra from the unit cube $[0, 1]^3$, leaving a central polyhedron whose volume fraction is $\frac{1}{4}$.